

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q15-0555

Thomas Hill Station 230 kV  
July 01, 2017

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0555. The Interconnection Customer proposes a 192.8 MW solar photovoltaic facility interconnecting at the Thomas Hill Station 230 kV substation in the Indiana area. The request was received on December 21, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 244 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,106,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

BLUEBONNET-JONES CHAPEL 230kV: 195.1% loading (758 MVA) following the outage of the ROBISON PARK-RUSSELL FORK 230kV; DFAX = 5.3%.

TWIN COUNTY RANCH-RICHLAND 230kV: 110.1% loading (649 MVA) following the outage of the TREGO-SWAN VALLEY 500kV Line; DFAX = 9.3%.

For the loss of DEKALB-JONES CHAPEL 500kV, the ROBISON PARK-RUSSELL FORK 230kV loads to 109.1% of its 1,695 MVA rating. Project DFAX: 6.3%.

For the loss of SWAN VLY-TREGO 500kV line, the STATE STREET-ROBISON PARK 230kV line loads to 105.7% of its 1,330 MVA rating. Project DFAX: 15.4%.

CONCORD-BLUEBONNET 230kV Line: 104.1% loading (1,311 MVA) following the outage of the JONES CHAPEL-DEKALB 500kV; DFAX = 5.1%.

WAYNEDEALE-MONFORT 230kV: 103.4% loading (1,112 MVA) following the outage of the MONTPELIER ELECTRIC GENERATING STA-WABSH 500kV; DFAX = 9.8%.

WAYNEDEALE-STATE ST 230kV line: 103.4% loading (833 MVA) following the outage of the MUSKOGEE-MONTPELIER ELECTRIC GENERATING STA 500kV L; DFAX = 12.4%.

The DOOVILLE-THOMAS HILL STATION 230kV is overloaded to 101.3% of its 578 MVA emergency rating for the loss of the WABASH SUBSTATION-MONTPELIER ELECTRIC GENERATING STA 500kV. The project contributes 23.9% DFAX (46.1 MW) to this overload.

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

#### 4. Required Network Upgrades and Cost Estimates

| Facility                             | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------|---------------------------------------------|---------------|-----------------------|
| BLEBONNET-JONES CHAPEL 230kV         | Rebuild line with higher-capacity conductor | \$31,042,000  | 1% (\$255,000)        |
| TWIN COUNTY RANCH-RICHLAND 230kV     | Rebuild line with higher-capacity conductor | \$29,641,000  | 3% (\$1,001,000)      |
| ROBISON PARK-RUSSELL FORK 230kV      | Reconductor line                            | \$17,759,000  | 1% (\$128,000)        |
| STATE STREET-ROBISON PARK 230kV line | Reconductor line                            | \$3,361,000   | 2% (\$67,000)         |
| CONCORD-BLEBONNET 230kV L            | Reconductor line                            | \$9,005,000   | 1% (\$73,000)         |
| MONFORT-WAYNE DALE 230kV             | Reconductor line                            | \$21,047,000  | 2% (\$463,000)        |
| WAYNE DALE-STATE STREET 230kV Line   | Reconductor line                            | \$3,827,000   | 3% (\$121,000)        |
| DOOVILLE-THOMAS HILL STA 230kV line  | Reconductor line                            | \$17,537,000  | 11% (\$1,998,000)     |
|                                      | Total allocated to project                  |               | \$4,106,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.